

The Darwinian revolution challenged traditional views of a young Earth inhabited by unchanging species

More than a century and a half ago, Charles Darwin was inspired to develop a scientific explanation for the *diversity of life*, the great number and remarkable variety of species on Earth. When he published his hypothesis in his book *The Origin of Species*, Darwin ushered in a scientific revolution—the era of evolutionary biology. As we’ll see, Darwin developed his revolutionary ideas over time, influenced by the work of others and his travels (Figure 22.2).

Endless Forms Most Beautiful

To set the stage for our study of Darwin and evolutionary biology, let’s return to the well-hidden orchid mantis in Figure 22.1. This species is a member of a diverse group of

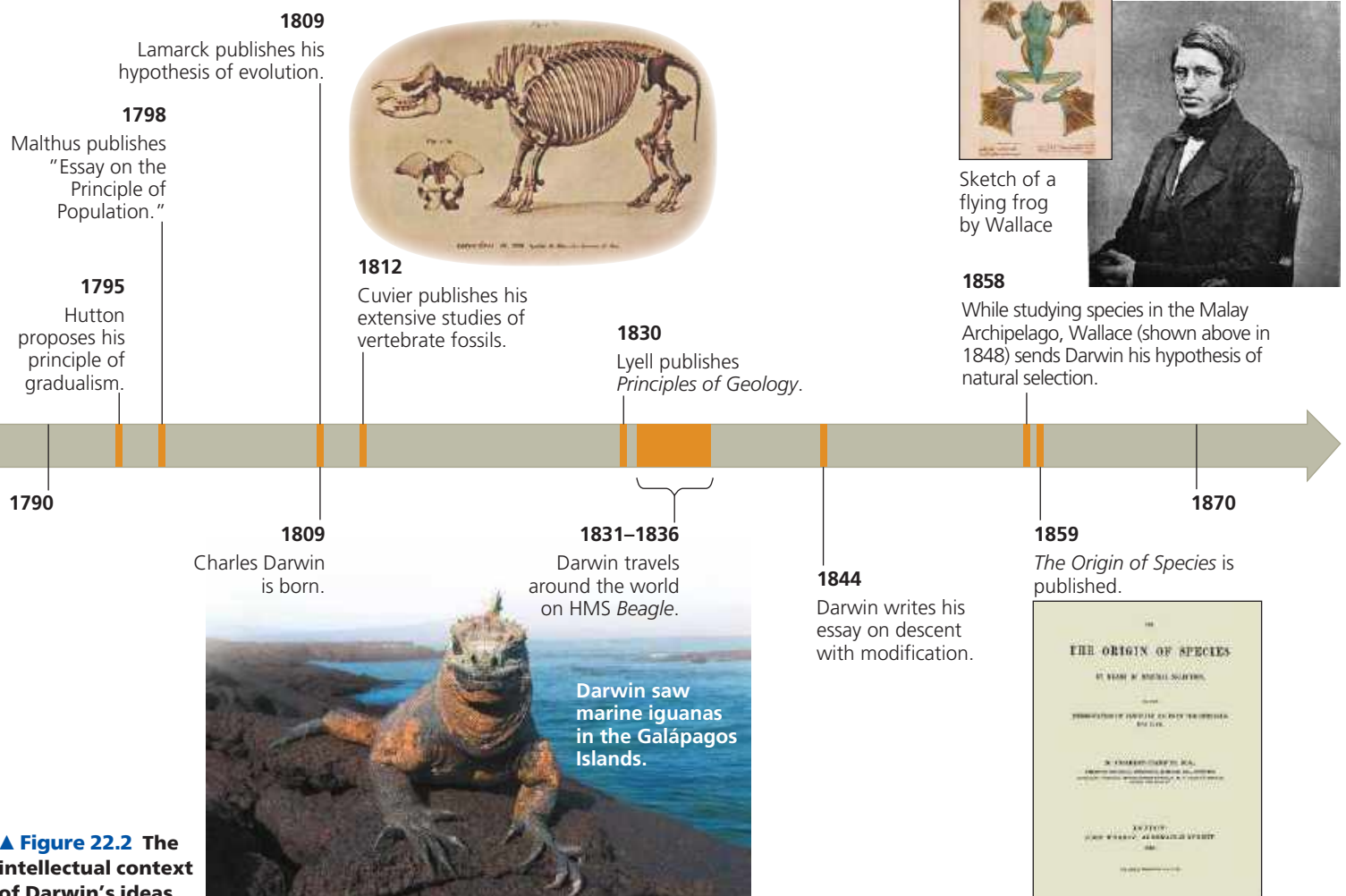
insects, the Mantodea, which includes 2,300 species in 430 genera. All these mantises share certain features, such as three pairs of legs, triangular heads with bulging eyes, and a flexible “neck.” Such shared features illustrate the unity of life, a phrase that highlights the fact that all organisms share characteristics.

But Earth’s many different species also differ from one another. The mantises, for example, differ in features such as their size, shape, and color. Overall, the orchid mantis and its many close relatives illustrate three key observations about life:

- organisms are well suited (adapted) for life in their environments (Here and throughout this text, the term *environment* refers to other organisms as well as to the physical aspects of an organism’s surroundings.)
- the many shared characteristics (unity) of life
- the rich diversity of life

Darwin, a keen observer of nature since childhood, set out to explain these three broad observations—an effort that eventually led him to conclude that life evolves over time.

For now, we will define **evolution** as *descent with modification*, a phrase Darwin used to summarize the process



▲ **Figure 22.2** The intellectual context of Darwin’s ideas.